

Discovery Report: Manil Doudou's Internship at Cavalons



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1. Introduction and Context of the Internship

This document constitutes the discovery report for my third-year engineering internship, a mandatory component of the FISE programme at CESI Toulouse. The primary objective of this report is to provide a formal analysis of my initial integration into the host company, **Cavalons**, examining its organizational structure, operational methodologies, and corporate culture. Furthermore, it serves as a reflective exercise to evaluate the alignment between my academic training in computer science and its practical application within a professional environment in France.

The internship is specifically focused on contributing to the development of **ÉquiProgress AI**, an AI-driven coaching chatbot that represents Cavalons' strategic expansion from its foundational peer-to-peer connection platform. My assigned responsibilities encompass several key areas: the design and implementation of conversational flows using the Botpress platform, integration with external AI services via APIs, analysis of equestrian datasets to inform the best recommendations, and the establishment of testing protocols to ensure response safety and contextual relevance.

This engagement provides a critical junction between theoretical software engineering principles and their implementation within a dynamic startup ecosystem. It offers substantive exposure to full-stack development processes, the integration of large language models (LLMs), agile project management practices, and the holistic product development lifecycle. The subsequent sections of this report will provide a detailed narrative of my daily activities, technical contributions, and analytical observations regarding the company's strategic direction and operational efficacy.

2. Company Presentation: Cavalons and the Equestrian Sector



Cavalons is a French startup founded in 2023 by entrepreneur Camille Despesse & developer Théo Cousinet. The company was established to address pronounced accessibility and affordability barriers within the equestrian sector. Its core business

model operates a digital platform that facilitates connections between horse owners and riders through three principal service offerings: **demi-pension** (structured long-term horse sharing), **horse sitting** (temporary care arrangements), and **experience sharing** (including guided lessons and trail rides). This platform leverages technology to reduce participation costs, optimize resource utilization, and foster trust-based community interactions.

The company operates within a substantial yet traditionally structured market. The French equestrian sector comprises approximately **2.5 million equids** and **2.7 million active riders**, generating an estimated €11 billion in annual economic activity. Despite this scale, access remains largely mediated through conventional equestrian centres, which involve significant financial commitment. Cavalons positions itself at the convergence of the **sharing economy** and specialised leisure services, utilizing digital infrastructure to create a more flexible and economically accessible alternative.

Cavalons is currently undergoing a strategic evolution with the development of their new application: **ÉquiProgress**. This initiative marks a transition from a purely connective platform to a comprehensive digital ecosystem. **ÉquiProgress** is conceived as an AI-powered coaching assistant that provides personalized training regimens, tracks equine wellness, and monitors user progression. This product extension aims to address secondary pain points related to training guidance and progress tracking, thereby deepening user engagement and unlocking new revenue streams. The company maintains a lean operational structure, consisting of two full-time members (the CEO and Lead Developer) supplemented by my internship role, enabling agile responsiveness and rapid iterative development.

3. Daily Internship Experience and Integration

3.1. Onboarding and Initial Integration

The onboarding process at Cavalons was notably efficient and immersive. The initial day involved a comprehensive briefing with the CEO, Camille Despesse, covering the company's mission, market analysis, competitive landscape, and strategic roadmap. This was immediately followed by a technical orientation session with the Lead Developer, Théo Cousinet, which provided an overview of the existing technology stack (React Native, etc), the architecture of the new **ÉquiProgress** project, and the collaborative tools in use (Trello). This dual-focus introduction successfully established both the commercial context and technical foundation necessary for my contribution.

Practical integration was accelerated through the immediate assignment of meaningful tasks. Within the first 48 hours, I was conducting analyses of anonymized equestrian training datasets to identify patterns for the AI knowledge base. By the conclusion of the first week, I had established my local development environment, familiarized myself

with the codebase, and begun constructing initial dialogue flows within the Botpress platform. This approach of "*learning through contribution*" demonstrated the company's philosophy of treating interns as capable team members from the outset.

3.2. Daily Rhythm and Operational Structure

The daily workflow follows a consistent pattern designed to optimize both individual productivity and team synergy. The day typically begins at 10:00, beginning with a period of individual planning and task prioritization. Morning hours are generally dedicated to focused development work, such as implementing specific chatbot functionalities, refining natural language processing (NLP) configurations, or analyzing user test feedback. I also use this time to review what I've done the day before and see if nothing is missing so I can do the further work in the afternoon after the lunch break.

Afternoon sessions continue development efforts but frequently incorporate collaborative problem-solving, especially when addressing integration challenges between the chatbot, backend services, and external APIs depending on the work.

A cornerstone of daily operations is the **team stand-up meeting at 17:30**. This 30–45-minute session serves multiple critical functions: reviewing progress on daily objectives, identifying and addressing immediate technical or conceptual blockers, aligning priorities for the following day, and periodically discussing broader strategic or architectural questions. This ritual ensures continuous alignment within the small team and maintains a clear, adaptive trajectory for the project.

3.3. Tools and Work Environment

The physical workspace is configured to promote collaboration, featuring an open-plan layout that facilitates spontaneous discussion and joint problem-solving. From a technological perspective, the development stack is contemporary and cloud-centric:

- **Conversational AI:** Botpress for dialogue management and NLP development.
- **Backend & Database:** Supabase (Tables, Cloud Functions, Authentication).
- **Frontend:** React Native for cross-platform mobile application development.
- **Communication & Project Management:** Trello & Google meetings.

I have access to all relevant repositories, documentation, datasets and project management systems that I need from day one, minimizing barriers to autonomous productivity within the established collaborative framework of the company.

4. The ÉquiProgress Project: AI Chatbot Development

4.1. Project Overview and Assigned Responsibilities

The ÉquiProgress project represents Cavalons' strategic initiative to develop an intelligent, context-aware digital coach for the equestrian community. The chatbot is designed to synthesize multiple data streams (including rider objectives, equine health history, past training sessions, and scheduled veterinary care) to generate personalized exercise recommendations, track progress, and provide educational guidance.

My primary technical responsibility has been the design, implementation, and refinement of the chatbot's conversational architecture using the **Botpress** platform. This encompasses creating intuitive dialogue flows, configuring intent recognition and entity extraction, integrating the Gemini API for response generation, and implementing the core business logic that ensures recommendations are safe, pedagogically sound, and tailored to the individual user-horse pair.

4.2. Specific Development Tasks and Cycle

A standard development cycle is initiated by analysing user stories or feedback to define required conversational pathways. For instance, a fundamental user query such as *"What exercise should I do today with my horse?"* necessitated designing a flow that systematically gathers contextual data: recent training intensity, horse wellness indicators, the rider's primary goals, and available session duration.

The technical execution involves:

1. Defining intent patterns within Botpress to correctly identify the user's request.
2. Designing entities to extract key variables (e.g., horse name, discipline, energy)
3. Structuring the dialogue to collect necessary context through natural prompts.
4. Configuring the integration and ensuring the prompt provides sufficient structured data (user profile, horse profile, history) for a contextual response.

4.3. Knowledge Base Curation and Safety Protocols

A significant portion of my work has involved structuring the domain-specific knowledge that grounds the AI's recommendations. This has entailed:

- Researching and synthesizing training methodologies from established equestrian federations (especially the French Equestrian Federation)
- Compiling a categorized library of exercises, annotated with discipline, difficulty, objectives, and common pitfalls through datasets (.csv files) & relevant websites that I converted to Word docs & also some pdfs with high quality relevant data.

- I've oriented the chatbot to exclusive equestrian answers so the chatbot isn't able to respond to other questions but the only the ones related to horse riding.

This knowledge is organized in a layered architecture within the system, allowing the AI to reference fundamental principles, specific exercise details, and safety rules.

Crucially, this structured knowledge also informs the **guardrail system**, a set of programmatic rules that pre-filter user queries, constrain AI suggestions in high-risk scenarios, and mandate professional consultation for specific health-related topics.

4.4. Testing and Quality Assurance Framework

Quality assurance is a multi-phase; rigorous process integrated into the development lifecycle. Each new conversational flow undergoes:

- **Functional Testing:** Verifying the technical execution of the dialogue with the AI chatbot & after that the integration to the application via API key implementation
- **Scenario Testing:** Evaluating the chatbot's responses across a wide range of simulated user inputs and contexts.
- **Safety Testing:** A critical phase where responses are scrutinized for potential risks, inaccuracies, or inappropriate recommendation.

Testing is both manual (leveraging the team's equestrian knowledge) and automated, using scripts to check for response consistency. Particular emphasis is placed on scenarios where the chatbot must correctly identify its own limitations and defer to professional veterinary or coaching expertise.

5. Technical Work and Methodologies

5.1. Development Stack and Implementation Details

My technical work utilizes a specialized stack focused on conversational AI:

- **Botpress:** Serves as the central orchestration platform, handling natural language understanding (NLU), dialogue state management, and integration connectors including the AI model working in it.
- **Supabase Backend:** Firestore stores persistent user profiles, conversation histories, and horse data. Cloud Functions host serverless logic for calculating training loads, analyzing activity history, and other computationally discrete tasks I developed.

5.2. Data Analysis Contributions

Beyond core development, I have conducted significant data analysis to inform product strategy and algorithmic logic. Using **Python** with **pandas** and **matplotlib**, I have processed internal datasets to:

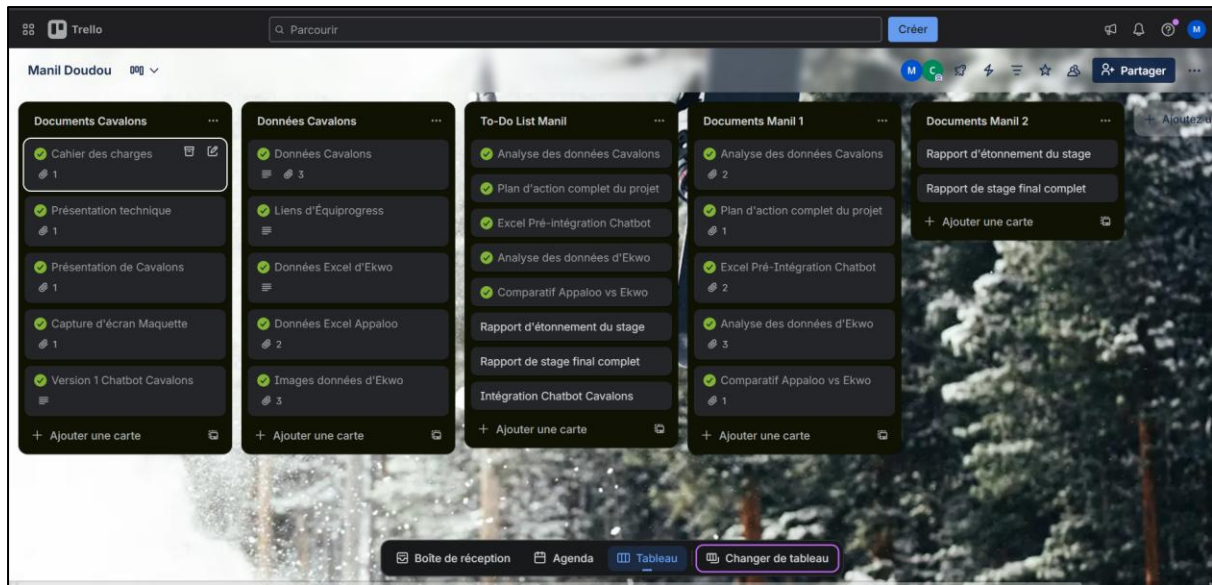
- Identify patterns in training frequency and intensity distribution.
- Analyse correlations between specific exercise types and user-reported outcomes or progress.
- Examine common progression bottlenecks for riders at different skill levels.

One impactful analysis focused on **post-treatment recovery patterns**. By analyzing data on training resumption after veterinary interventions (e.g., injections, osteopathy), I helped establish evidence-based convalescence guidelines that are now embedded in the chatbot's logic, ensuring recommended workloads are appropriately reduced during recovery periods.

5.3. Agile-Informed Development Practices

The development methodology embraces agile principles in a pragmatic; lightweight form tailored to a startup environment. Rather than following rigid, time-boxed sprints, work advances in flexible cycles focused on delivering concrete features. Team coordination is maintained through daily stand-ups and weekly planning sessions, ensuring regular synchronization without excessive administrative overhead.

Task and document management is centralized using **Trello**, which serves as the primary organizational hub. The Trello board is structured with workflow columns (Backlog, In Progress, Testing, Done), with each card representing a specific feature, conversational flow, or document deliverable. This platform is used to organize my daily tasks, track progress on assigned work, and manage document flow—both for receiving project specifications and datasets from the team, and for submitting completed deliverables. At any given time, my focus is intentionally limited to three to five active cards, balancing the development of new functionality with iterative refinement of existing features based on testing feedback. This iterative, feedback-driven approach is particularly well-suited to AI development, where effective behaviour emerges progressively through continuous experimentation and adjustment.



6. Team Collaboration and Company Culture

6.1. Collaborative Dynamics and Role Integration

Collaboration with Camille (CEO) and Théo (Lead Developer) is characterized by a fluid, cross-functional dynamic. Camille provides strategic direction, user insight, and defines product requirements based on market understanding. Théo offers architectural guidance, oversees backend integrations, and ensures system scalability. My role operates at the intersection, translating product requirements into functional conversational logic and ensuring technical implementation aligns with user experience goals, and I really hope that I'm providing real value to the Cavalons's workflow

Collaboration modes shift based on task nature:

- **Technical Implementation:** Involves close coordination with Théo on API design, data models, and performance optimization.
- **Product Design:** Engages more frequently with Camille to review conversation scripts, safety protocols, and educational value.
- **Strategic Alignment:** Often involves all three parties in discussions to ensure technical decisions support business objectives and user needs.

The working relationship is built on **mutual respect, intellectual curiosity, and a shared commitment to project quality**. Both Camille and Théo consistently demonstrate patience in knowledge transfer, openness to alternative solutions, and genuine investment in my professional growth and I really appreciate this.

6.2. Communication Protocols and Meeting Structures

The **daily 17:30 meeting** is the operational keystone. Its consistent structure (progress sharing, blocker identification, priority alignment) ensures team cohesion and momentum. Approximately weekly, this meeting expands to include deeper strategic or technical architecture reviews. Additional, ad-hoc "sync" sessions occur spontaneously when specific integration challenges require focused problem-solving.

6.3. Embodiment of Corporate Values

The company's stated values of **Sharing, Trust, and Safety** are authentically enacted in daily operations:

- **Sharing:** Manifested in the open exchange of knowledge, collaborative debugging sessions, and collective ownership of both challenges and successes. Information is never siloed.
- **Trust:** Forms the foundation of team dynamics, enabling significant autonomy with minimal oversight. This trust is reinforced by a culture of voluntary transparency and reliability.
- **Safety:** This is the paramount concern, dual-applied to both product and workplace. For the chatbot, it translates into rigorous safety protocols and conservative AI guardrails. Internally, it fosters **psychological safety**, where admitting uncertainty, asking questions, and proposing unconventional ideas are encouraged without fear of negative judgment.

7. Learning Outcomes and Skill Development

7.1. Technical Skill Acquisition

The internship has substantially advanced my technical proficiencies:

- **Conversational AI & NLP:** Gained practical expertise with the Botpress platform, including intent/entity configuration, dialogue management, and integration with external LLM APIs.
- **Full-Stack Development:** Enhanced understanding of integrating frontend conversational interfaces with Supabase backend services including data modelling and serverless logic implementation.
- **Data Analysis:** Applied Python data science libraries (pandas, matplotlib) to derive actionable insights from real-world equestrian data, directly informing product logic, visuals and insights

7.2. Professional Competency Development

Beyond technical skills, key professional competencies were cultivated:

- **Cross-Functional Communication:** Improved ability to articulate technical concepts to non-technical stakeholders (CEO) and to translate business requirements into technical specifications.
- **Systematic Problem-Solving:** Matured my approach to debugging complex systems and decomposing large, ambiguous problems into actionable steps.
- **Quality Assurance Mindset:** Developed a rigorous attitude towards testing, particularly regarding safety-critical AI outputs, understanding that deployment requires more than just functional correctness.

7.3. Domain Knowledge and Business Acumen

- **Equestrian Domain Literacy:** Acquired foundational knowledge of equestrian terminology, training principles, biomechanics, and welfare considerations, which is essential for developing a credible product in this space.
- **Startup Operations:** Gained firsthand insight into the realities of a bootstrapped startup, including resource prioritization, the interplay between technical debt and feature velocity, and product-led growth strategies.
- **Business-Technology Intersection:** Observed how technical architecture decisions directly impact business metrics like development speed, user satisfaction, and operational cost, fostering a more holistic view of engineering impact on the end result.

8. Challenges and Problem-Solving

8.1. Technical and Integration Challenges

- **AI Safety and Relevance:** Ensuring the chatbot generated consistently safe and appropriate recommendations was an initial challenge. The solution involved implementing a multi-layered approach: pre-processing filters to flag high-risk queries, carefully engineered system prompts with explicit safety instructions, and post-processing validation rules.
- **Conversation Context Management:** Maintaining coherent context across multi-turn dialogues where users provide information piecemeal required refining Botpress's context management and designing flows that strategically confirm key details without being repetitive.
- **System Latency:** Early integration patterns resulted in slow response times due to sequential API calls. This was optimized by redesigning data flows,

implementing client-side caching of static user/horse data, and restructuring dialogues to be more efficient in gathering necessary context.

8.2. Methodological and Conceptual Adaptation

- **Transition to Agile Fluidity:** Moving from academia's defined-specification projects to an agile environment required a mindset shift. I adapted by embracing iterative development, learning to define "minimum viable" features for testing, and valuing user feedback as a primary driver for improvement.
- **Balancing Innovation with Constraint:** Navigating the tension between creating a personalized, innovative AI coach and enforcing strict safety guardrails was a constant negotiation. The evolved "risk-tiered" model, which varies AI autonomy based on scenario risk, emerged as a balanced solution, allowing creativity in safe contexts while enforcing strict rules in others.

9. Strategic Observations and Recommendations

9.1. Analysis of Current Operations

Cavalons demonstrates notable strengths in **agile execution, collaborative culture, and strategic vision**. The lean structure allows for rapid decision-making and iteration. The focus on ÉquiProgress represents a good product expansion that leverages existing platform assets to address user needs based on direction of the company itself.

Observations on Operational Scale:

1. Knowledge Preservation

The team's current reliance on verbal exchanges and unrecorded decisions is highly efficient for a compact, co-located team. However, this model depends heavily on institutional memory held by a few individuals. As the organization scales, this approach may create bottlenecks in onboarding and increase the risk of knowledge loss or misalignment, suggesting a future need to balance conversational efficiency with more systematic documentation.

2. Process Formalization

Existing workflows are characterized by organic, adaptable collaboration with minimal formal procedure. This agility is a current strength that enables rapid iteration. An observation is that this informality, while effective at the present scale, may require incremental structuring to maintain consistent quality and coordination as the team expands, without introducing unnecessary bureaucracy.

3. Testing Infrastructure

Quality assurance currently hinges on manual testing, which aligns with the project's early-stage complexity and allows for nuanced evaluation of AI responses. It is

observable that the codebase's growing size and feature count will naturally increase the testing burden. This growth presents a future consideration for integrating automated checks to safeguard existing functionality, thereby allowing manual efforts to focus on complex, high-value test scenarios.

9.2. Recommendations

1. Start a "Decision Journal"

- Problem: Important "why" decisions about the product and code are only talked about, not written down. This works now with a small team but will cause confusion and repeated mistakes when more people join.
- Solution: Create a simple shared document. Every time the team makes a key decision, write down what was decided and the main reason why. This saves time later when anyone needs to remember the original thinking.

2. Agree on a Few Simple Work Rules

- Problem: As the team expands, people might work in different ways. Without basic quality checks, small bugs can slip through, and what counts as a "finished task" might become unclear.
- Solution: Set 2-3 easy rules to keep work consistent and high-quality. For example: 1) All code gets a quick look from another teammate before it's finalized. 2) Every task has a mini checklist (like "code, test, document") so everyone agrees when it's truly done.

3. Build a Small "Safety Net" of Automated Tests

- Problem: The team manually tests everything. As the chatbot gets more features, this takes longer and it's easier to miss a bug that breaks an existing function.
- Solution: Don't test everything automatically at once. Start by writing a simple script that automatically checks the 3-5 most important chatbot conversations every day. This acts as a "safety net" to catch major breaks immediately, saving time for more complex testing.

10. Conclusion and Personal Reflection

This discovery report has detailed my successful integration and substantive contributions during the initial phase of my engineering internship at Cavalons. The experience has been marked by a seamless onboarding process, immersion in a positive and values-driven culture, and engagement with technically challenging and meaningful work on the ÉquiProgress AI chatbot.

Professionally, this internship has fundamentally broadened my understanding of software engineering. I have moved beyond viewing it purely as a technical implementation discipline to appreciating its role as a strategic, user-centred, and ethically grounded practice. The necessity of deep domain understanding for effective innovation, the constant negotiation between technical possibilities and business constraints, and the paramount importance of safety in AI applications are lessons that will indelibly shape my future career especially as an AI engineer.

The Cavalons environment (characterized by trust, intellectual collaboration, and a mission-driven focus) has proven to be an ideal setting for applying academic knowledge to real-world problems. I am grateful to Camille Despesse and Théo Cousinet for their exceptional guidance and for fostering such a productive and supportive learning environment. This internship has not only advanced my technical skill set but has also provided a foundational template for the kind of impactful, conscientious engineering work I aspire to contribute to my professional future.

11. Acknowledgements

I would like to extend my sincere gratitude to **Camille Despesse**, CEO of Cavalons, for her exceptional leadership, for entrusting me with meaningful responsibilities from the outset, and for her unwavering support and insightful guidance throughout this internship. Her clear strategic vision and deep understanding of the equestrian world were invaluable in shaping my contribution to the ÉquiProgress project. My heartfelt thanks also go to **Théo Cousinet**, Lead Developer, for his technical expertise, immense patience, and collaborative spirit. His willingness to explain complex concepts, review my work, and engage in productive problem-solving sessions was fundamental to my technical learning and the successful completion of my assigned tasks. The welcoming, trusting, and intellectually stimulating environment they fostered together made this internship an incredibly enriching and formative professional experience.